

Solution Requirements

SMARTBRIDGE / SKILL WALL

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Project: EduGenie: Google Gemini Powered Learning Assistant

Define Functional and Non-Functional Requirements:

FR = what the system should do. NFR = how well the system should operate.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	No login is required in the current MVP. The learning interface is directly accessible to students.	
2	Authorization levels	Single student/learner role. Users can access all learning tasks; no admin role is required in the MVP.	
3	External interfaces	FastAPI REST endpoints connect the web UI with Google Gemini. Local LaMini-Flan-T5-783M supports explanation.	

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S.No	Requirement Category	Requirement Description	Priority
4	Transactions processing	Student requests are validated, routed to the selected task, processed by the AI module, and returned as a learning result.	High
5	Reporting	The system presents answers, summaries, quiz results, explanations, and learning recommendations. No external export is required in the MVP.	Medium
6	Business rules, etc.	Empty inputs are rejected; the selected task determines the module type. Quizzes use structured MCQs; learning paths use three levels.	High
7	Compliance to laws or regulations	Protect the Gemini API key on the backend using environment configuration. Avoid unnecessary student data collection.	Medium
8	Other	Provide clear validation and error messages, maintain separate API endpoints, and expose core learning functions through REST APIs.	Medium

Step 2: Non-Functional Requirements (NFR)

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S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Health endpoint should respond quickly; AI tasks should complete within 60 seconds. AI tasks complete without timeout under normal conditions.	
2	Scalability	Keep FastAPI endpoints and AI modules separated so additional AI models or processing components can be added independently.	
3	Security & Data Privacy	Keep Gemini API key server-side and never expose secrets in HTML or JavaScript on frontend. GitHub validate inputs.	
4	Reliability & Availability	Return clear validation errors and handle model/API failures with out-of-the-box application return a controlled error.	
5	Usability & Accessibility	Provide a simple responsive HTML/CSS interface with clear task definitions and readable results. Interfaces and readable results and results are clearly displayed.	
6	Other	Use modular Python files and automated tests for core API/modules. Core modules remain independently testable; tests run before submission.	